

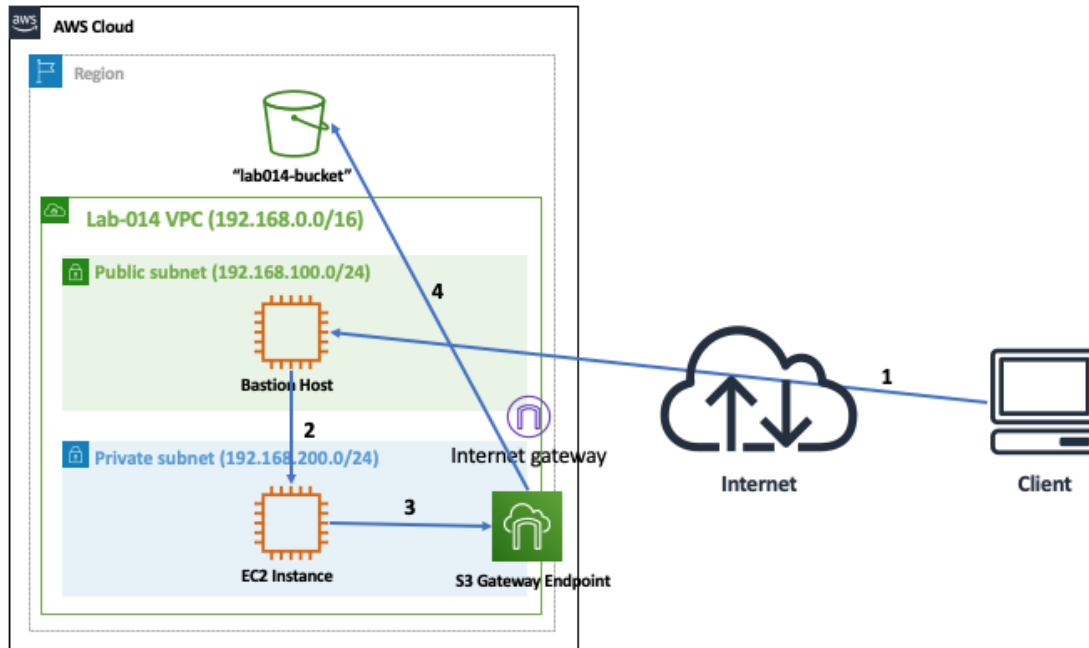
Lab-014

An S3 Bucket Accessed via an S3 Gateway Endpoint

Goal

The goal of this lab is to illustrate how to access an S3 bucket from an EC2 instance in a private subnet using an S3 gateway endpoint.

Architecture Diagram



Overview

The architecture diagram illustrates the steps we will have to take to demonstrate how the EC2 instance in the private subnet can still access an S3 bucket even if the instance does not have access to the internet.

1. From the client you can connect to a bastion host configured in the public subnet (see lab-02),
2. From the bastion host you can then connect to the EC2 instance in the private subnet,
3. From the EC2 instance you can use the S3 gateway endpoint, to ...
4. Access the S3 bucket.

Step 1 - Initial Setup

Follow the steps described in lab-02, making sure you change all references to lab-002 with lab-014.

Step 2 - Create an S3 Bucket

Name your S3 bucket *lab014-bucket*. Refer to lab-13 if not sure how to do this step.

Step 3 - Create an S3 Gateway Endpoint

Go to VPC - Endpoints and click on *Create Endpoint*.

Create Endpoint

Actions ▾

Filter by tags and attributes or search by keyword

You do not have any Endpoints in this region

Click the Create Endpoint button to create your first Endpoint

Create Endpoint

[Endpoints](#) > Create Endpoint

Create Endpoint

A VPC endpoint allows you to securely connect your VPC to another service.
An interface endpoint is powered by [PrivateLink](#), and uses an elastic network interface (ENI) as an entry point for traffic destined to the service.
A gateway endpoint serves as a target for a route in your route table for traffic destined for the service.

- Service category**
- ☒ AWS services
 - ☐ Find service by name
 - ☐ Your AWS Marketplace services

Service Name com.amazonaws.us-west-1.s3 ⓘ

search : s3 add filter

<< < 1 to 1 of 1 > >>

Service Name	Owner	Type
☒ com.amazonaws.us-west-1.s3	amazon	Gateway

VPC* vpc-080c2ae1ce3f75272 ↕ ⓘ

Configure route tables A rule with destination **pl-6ba54002 (com.amazonaws.us-west-1.s3)** and a target with this endpoints' ID (e.g. vpce-12345678) will be added to the route tables you select below.

Subnets associated with selected route tables will be able to access this endpoint.

rtb-03ea4195fc0e60b5d ⓘ

Route Table ID	Main	Associated With
☒ rtb-03ea4195fc0e60b5d	Yes	2 subnets

- Policy*** ☒ Full Access - Allow access by any user or service within the VPC using credentials from any AWS accounts to any resources in this AWS service. All policies — IAM user policies, VPC endpoint policies, and AWS service-specific policies (e.g. Amazon S3 bucket policies, any S3 ACL policies) — must grant the necessary permissions for access to succeed. ?
- ☐ Custom

Use the [policy creation tool](#) to generate a policy, then paste the generated policy below.

```
{
  "Statement": [
    {
      "Action": "*",
      "Effect": "Allow",
      "Resource": "*",
      "Principal": "*"
    }
  ]
}
```

Key	Value
(128 characters maximum)	(256 characters maximum)

This resource currently has no tags

Add Tag 50 remaining (Up to 50 tags maximum)

* Required

[Cancel](#)

Create endpoint

Create Endpoint

✓ The following VPC Endpoint was created:

VPC Endpoint ID `vpce-06752d8599b389e04`

Close

Test and Validation

Connect to the EC2 instance in the private subnet. Configure AWS CLI with an access key. Create some test files. Then follow a test procedure similar to the one described in [lab-013](#).

First list all of your buckets using:

```
aws s3 ls
```

If you want to list the objects (and folders) of your *lab014-bucket* bucket try:

```
aws s3 ls s3://lab014-bucket
```

Try uploading some files using:

```
aws s3 cp file1.txt s3://lab014-bucket
```

To illustrate downloading a file you can try:

```
aws s3 cp s3://lab014-bucket/file1.txt file1-copied-back.txt
```